CLINICAL · EAR

Otosclerosis

Otosclerosis is a disease of the bony capsule of the inner ear, in which dense bone is remodelled into spongy bone that can lock the stapes in the oval window. The patient is usually a young adult with slowly worsening hearing and a perfectly normal drum; the cure, when it is needed, is one of the most satisfying operations in otology.

AT A GLANCE

- Young adult with progressive conductive loss and a normal drum.
- Rinne negative, As tympanogram, Carhart's notch, absent reflex.
- Choose a hearing aid or stapes surgery.
- Don't operate on an only hearing ear.
- Small risk of dead ear, about 1 in 200.
- Stapedotomy closes the gap to ≤10 dB in 80–95%; dead ear 0.4–2%; offer a hearing aid first.

1 CHAPTER ONE Overview

A young adult, a normal drum, and a hearing loss that keeps creeping on.

1.1 Definition

Otosclerosis, also called *otospongiosis*, is a primary disease of the otic capsule. Normal dense enchondral bone is replaced by vascular, spongy bone that later hardens again. When a focus reaches the oval window it fixes the stapes footplate and causes a conductive hearing loss; when it involves the cochlea it can add a sensorineural loss.

1.2 Epidemiology

The histological disease is far commoner than the clinical one. Lee quotes histological foci in 8 to 12% of white adults, but clinical otosclerosis in only 0.5 to 2%. It is less common in Black and Asian populations. Women are affected about

twice as often as men, and hearing loss usually begins between 20 and 30 years of age; onset before 10 or after 40 is unusual.

1.3 Why it matters

Otosclerosis is the classic cause of progressive conductive deafness behind an intact, normal-looking drum. It is entirely treatable, with a hearing aid or with surgery that restores hearing in nine patients out of ten. But the operation opens the inner ear, and a small number of patients lose the hearing in that ear altogether, so the decision to operate is a careful one.

2 CHAPTER TWO Pathology

A remodelling disease of the otic capsule.

2.1 The otic capsule

The bony labyrinth is formed from three layers: an outer periosteal layer, an inner endosteal layer, and a middle enchondral layer that normally stays dense and inert throughout life. Otosclerosis is a disease of that middle layer. Its favourite site is the *fissula ante fenestram*, a small cleft of fibrous tissue just in front of the oval window.

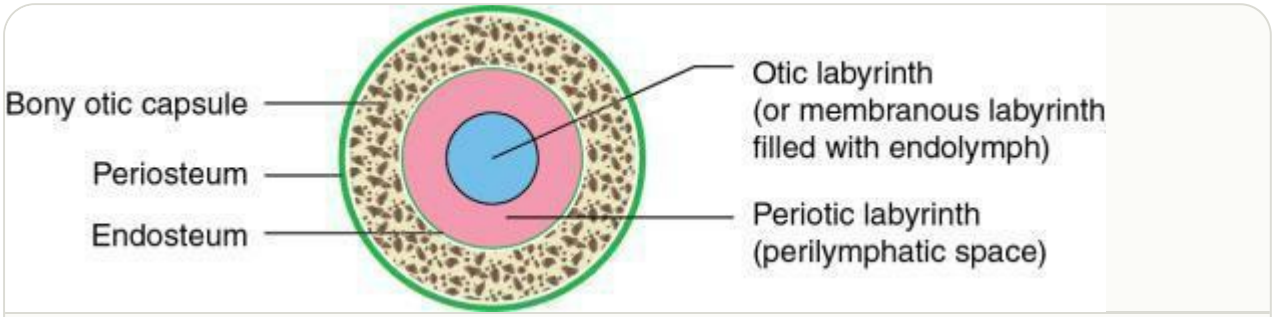


FIGURE 1 The bony and membranous labyrinth. The otosclerotic focus usually begins just in front of the oval window, where it can fix the stapes footplate.
Dhingra, fig. 13.1

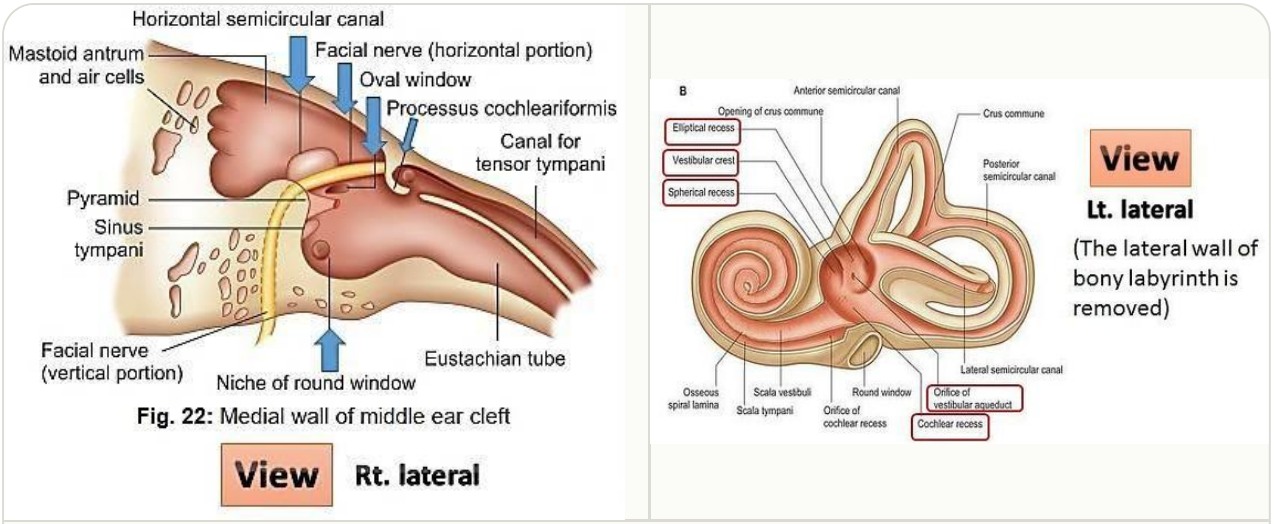


FIGURE 2 The medial wall of the middle ear with the oval window, and the bony labyrinth: the oval window niche is where stapedial otosclerosis fixes the footplate.

Anatomy atlas · ENT Steps p. 48, ENT Steps p. 66

2.2 Histology: active and mature foci

Grossly, a focus may look chalky white, grey or yellow; a red, vascular focus is active and likely to progress quickly.

2.2.1 Active (otospongiotic) phase

Immature lesions are full of marrow and vascular spaces, with osteoclasts resorbing bone and osteoblasts laying down new bone. Cement substance around the vessels stains blue with haematoxylin and eosin, forming the *blue mantles of Manasse*, the histological mark of activity.

2.2.2 Mature (sclerotic) phase

Older foci are less vascular, with more laminated bone and fibrillar matrix that stains red. Hearing loss reflects where the focus sits, not how active it is.

2.3 Types of otosclerosis

2.3.1 Stapedial otosclerosis

The commonest clinical form, causing stapes fixation and conductive loss. The focus may be:

- **Anterior:** at the fissula ante fenestram, the site of predilection.
- **Posterior:** behind the oval window.

- **Circumferential:** around the margin of the footplate.
- **Biscuit type:** within a thickened footplate, the annular ligament still free.
- **Obliterative:** filling the whole oval window niche.

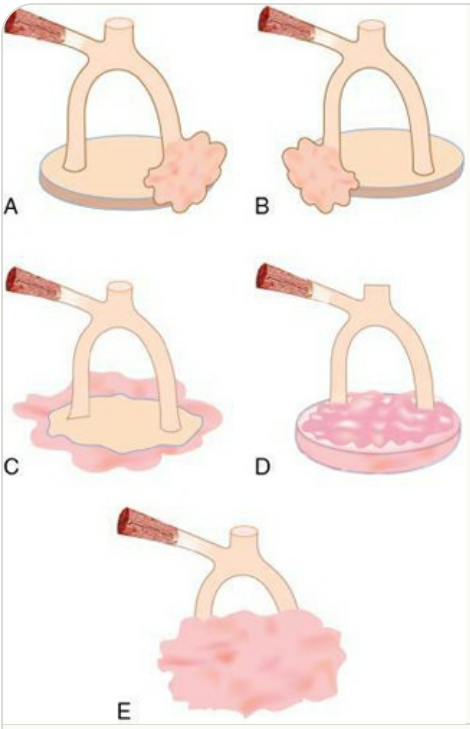


FIGURE 3 Types of stapedial otosclerosis: (A) anterior focus, (B) posterior focus, (C) circumferential, (D) biscuit type with a thick footplate, (E) obliterative.
Dhingra, fig. 13.2

2.3.2 Cochlear otosclerosis

Foci around the round window or elsewhere in the cochlear capsule cause sensorineural loss, probably from enzymes and toxic products released into the inner ear fluids. It is often combined with stapedial disease, giving a mixed loss.

2.3.3 Histological otosclerosis

Foci that touch neither the stapes nor the cochlea remain silent and are found only at post-mortem.

3 CHAPTER THREE Aetiology

Genes, hormones, and perhaps a virus.

3.1 Heredity

About half of patients have a family history. Inheritance is autosomal dominant with incomplete penetrance, and several genetic loci have been described.

3.2 Sex and hormones

Women are affected about twice as often, and the hearing loss may begin or accelerate during pregnancy, around the menopause, or after an accident or major operation.

3.3 Associations

3.3.1 Osteogenesis imperfecta

Brittle bones, blue sclerae and conductive deafness form *van der Hoeve syndrome*. The otic capsule lesions look identical to otosclerosis under the microscope, and both involve genes for type I collagen.

3.3.2 Measles virus

Measles RNA and antigens have been found in active foci, and some argue for a viral cause, as has been proposed for Paget's disease.

4 CHAPTER FOUR Clinical features

The history usually tells you; the drum does not.

4.1 Symptoms

4.1.1 Hearing loss

Painless, insidious and progressive, beginning in the twenties and usually affecting both ears, often one before the other.

4.1.2 Paracusis Willisii

The patient hears speech better in a noisy room. Background noise makes normal-hearing speakers raise their voices, while it barely reaches the otosclerotic ear.

4.1.3 Tinnitus and vertigo

Tinnitus is common, especially with cochlear involvement and active disease. Vertigo is uncommon.

4.1.4 Speech

Because the patient hears their own voice loudly by bone conduction, they speak softly, in a monotonous, well-modulated voice.

4.2 Signs

The tympanic membrane is normal and mobile, and Eustachian tube function is normal. In about one in ten, a pink blush shows through the drum over the promontory: *Schwartze sign*, which reflects increased vascularity in an active focus.

> **CLINICAL PEARL**
>
> A conductive loss with a normal drum has a short differential: otosclerosis, ossicular discontinuity, malleus head fixation, and superior canal dehiscence. The stapedial reflex separates the last of these (see
> 5.4
>).

5 CHAPTER FIVE Investigations

Forks, audiogram and impedance tell the story.

5.1 Tuning forks

Rinne turns negative first at 256 Hz, then at 512 Hz and finally at 1024 Hz as fixation progresses, so the forks give a rough measure of the air-bone gap. Weber lateralises to the ear with the greater conductive loss. Absolute bone conduction is normal, unless the cochlea is involved.

WHERE THE BOOKS DIFFER · AIR-BONE GAP THAT TURNS EACH FORK NEGATIVE

Dhingra	Lee
	15, 25 and 35 dB respectively.

256 Hz at about 15 dB, 512 Hz at about 30 dB, 1024 Hz at about 45 dB.

Both describe the same sequence: negative Rinne at 1024 Hz means a large gap.

5.2 Pure-tone audiometry

Air conduction is reduced, most at the lower frequencies early on. Bone conduction is normal, but often shows a dip at 2000 Hz: *Carhart's notch*, about 5 dB at 500 Hz, 10 dB at 1000 Hz, 15 dB at 2000 Hz and 5 dB at 4000 Hz. The notch is a mechanical artefact of stapes fixation, not a cochlear loss, and it disappears after successful surgery. A pure conductive loss rarely exceeds about 60 dB; anything worse means cochlear involvement.

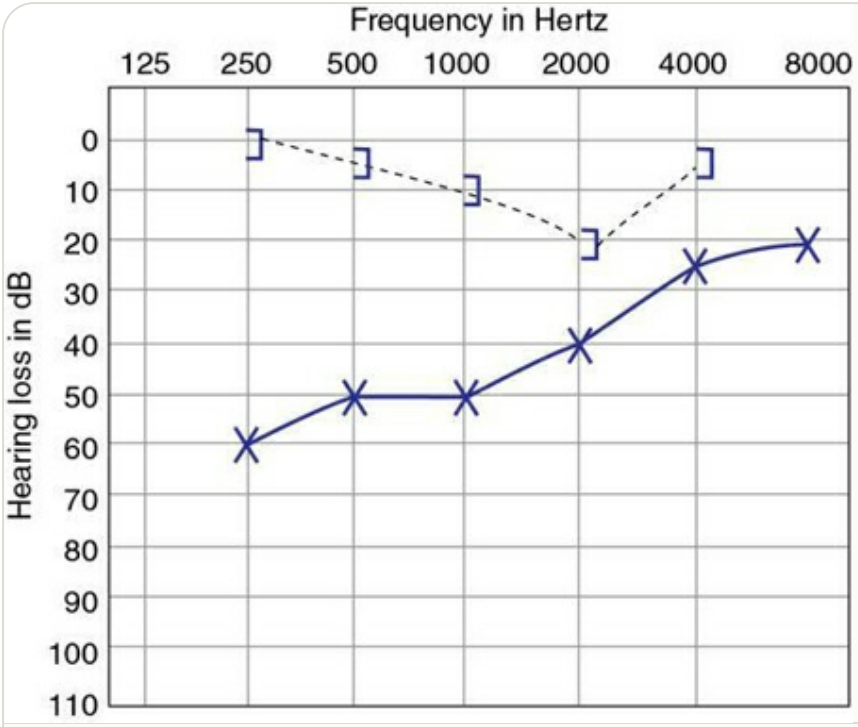


FIGURE 4 Audiogram of a left ear with otosclerosis. Air conduction is reduced with an air-bone gap; bone conduction dips at 2000 Hz (Carhart's notch).
Dhingra, fig. 13.3

5.3 Speech audiometry

Discrimination is good, unless there is cochlear disease. This matters for surgical selection.

5.4 Impedance audiometry

5.4.1 Tympanometry

Normal early; later a shallow, stiff As curve.

5.4.2 Stapedial reflex

Absent once the stapes is fixed. In early disease a characteristic biphasic (on-off) pattern may appear.

PEARL

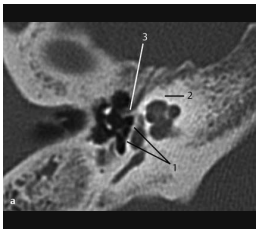
A conductive loss with a normal drum and a present stapedial reflex is not otosclerosis. Think of superior canal dehiscence, which produces an inner-ear conductive loss with intact reflexes.

5.5 Imaging

High-resolution CT is not needed routinely. It is useful when the loss progresses rapidly, when cochlear otosclerosis is suspected (a pericochlear halo, the *double ring sign*), when the gap is doubtful, before revision surgery, and when there are vestibular symptoms.

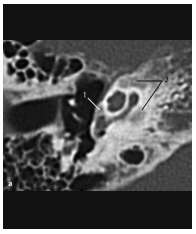
IMAGING IN THE APP 6 cases

CT signs of otosclerosis, a piston after stapedectomy, a gusher ear and osteogenesis imperfecta.



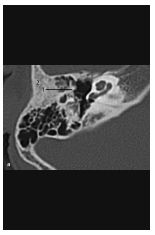
Otosclerosis, Radiologic Signs around the Oval Niche

Bone disease · 10 images



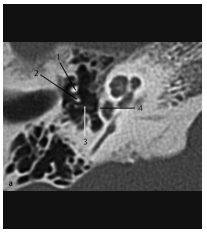
Otosclerosis, Retrofenestral

Bone disease · 3 images



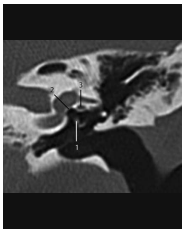
Fixation of the Ossicular Chain

Bone disease · 3 images



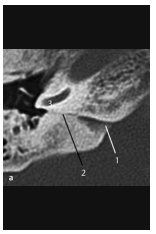
Stapedectomy, Control of Piston Position

After surgery · 4 images



Osteogenesis Imperfecta

Bone disease · 3 images



Gusher

Infection & inflammation · 3 images

CHAPTER SIX

6 Differential diagnosis

Other causes of a conductive loss.

6.1 Conductive loss with a normal drum

HOW TO TELL THEM APART

	CLUE	KEY TEST
Otosclerosis	Young adult, progressive, family history	Absent reflex, Carhart's notch
Superior canal dehiscence	Autophony, sound- or pressure-induced vertigo	Reflex present, low VEMP threshold, CT
Malleus head fixation	Similar history	Malleus immobile on palpation at surgery, CT
Ossicular discontinuity	Trauma or previous infection	Hypermobile drum, Ad tympanogram, gap often over 50 dB
Congenital stapes fixation	Childhood onset, non-progressive	Flat 50–60 dB loss, no Carhart's notch

6.2 Conductive loss with an abnormal drum

Otitis media with effusion, adhesive otitis media and tympanosclerosis are separated by otoscopy and tympanometry.

7 CHAPTER SEVEN Management

Watch, aid, or operate.

7.1 The three options

OTOSCLEROSIS

- MILD LOSS **Observation** review hearing yearly
- ANY LOSS **Hearing aid** effective, no risk; the choice for an only hearing ear
- SUITABLE GAP **Stapes surgery** closes the gap in 90–95%
- FAR ADVANCED **Cochlear implant** when aids no longer help

Each is legitimate. The patient chooses after an honest account of the benefits and the small risk of losing the hearing in the operated ear.

7.2 Medical treatment

No drug reverses fixation. *Sodium fluoride* has been used to hasten maturation of active foci and slow sensorineural progression in cochlear otosclerosis.

WHERE THE BOOKS DIFFER · SODIUM FLUORIDE

Dhingra	Lee
Controversial, and not generally recommended.	Widely accepted for progressive cochlear disease, though not FDA approved.

In practice it is considered for active disease with progressive sensorineural loss, a positive Schwartze sign or a halo on CT.

7.3 Hearing aid

An excellent alternative for patients who do not want surgery or are unfit for it, and the treatment of choice for an only hearing ear. Conductive losses aid particularly well.

7.4 Stapes surgery

The fixed stapes is removed, wholly or in part, and a prosthesis is placed between the long process of the incus and the inner ear.

7.4.1 Selection

The patient should feel the handicap (air conduction around 30 dB or worse), have a real air-bone gap with Rinne negative at 256 and 512 Hz, and keep good speech discrimination.

WHERE THE BOOKS DIFFER · CRITERIA FOR SURGERY

Dhingra	Lee
Air-bone gap at least 15 dB, discrimination 60% or more, ideal age 20–30.	Air-bone gap 20 dB or more (25–30 dB in the rehabilitation chapter), discrimination 70% or better, typical age 30–50.

Operate on the poorer-hearing ear first.

7.4.2 Contraindications

- **Only or better-hearing ear:** use a hearing aid.
- **Active Ménière's disease:** a hydropic ear is at higher risk of sensorineural loss.
- **Active middle ear disease or perforation:** treat first.
- **Children:** recurrent Eustachian tube problems and fast regrowth of the focus.
- **Occupations:** divers, pilots and frequent flyers, roofers and athletes (balance), and noisy workplaces, since the stapedius reflex is lost.
- **Relative:** otitis externa, exostoses and pregnancy; defer until settled.

7.4.3 Stapedectomy or stapedotomy

In *stapedectomy* all or part of the footplate is removed and the window sealed with a graft; in *stapedotomy* only a small fenestra (about 0.6–0.8 mm) is made in the footplate. Stapedotomy disturbs the inner ear less, gives better high-frequency results and has largely replaced stapedectomy.

7.4.4 Steps

The operation is usually done under local anaesthesia through the ear canal.

- Meatal incision and elevation of the tympanomeatal flap.
- Exposure of the oval window, removing the posterosuperior bony overhang with curette or drill, preserving the chorda tympani.
- Confirmation of stapes fixation and a mobile malleus and incus.
- Measurement from the incus to the footplate.
- Division of the stapedius tendon and removal of the stapes superstructure.
- A hole in the footplate (stapedotomy) or partial removal (stapedectomy).
- Placement of the prosthesis on the incus and into the fenestra, and crimping.
- Replacement of the flap.

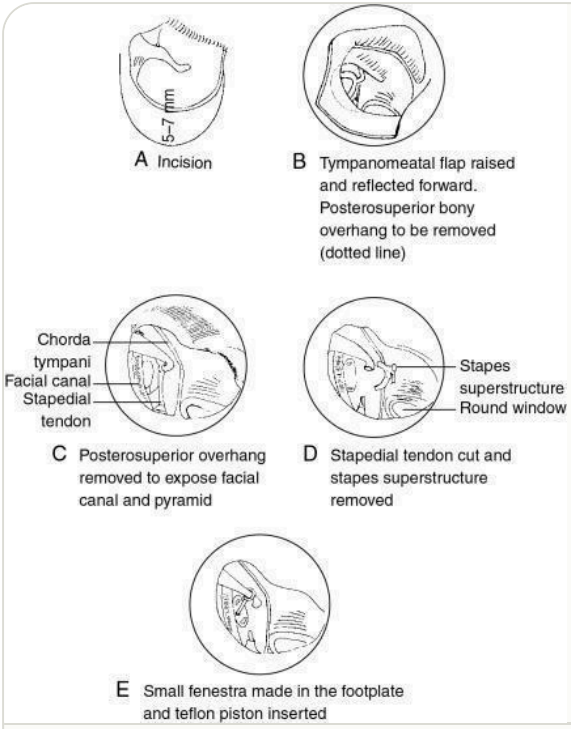


FIGURE 5 Steps of stapedectomy, from elevation of the tympanomeatal flap to placement of the prosthesis.

Dhingra, fig. 13.6

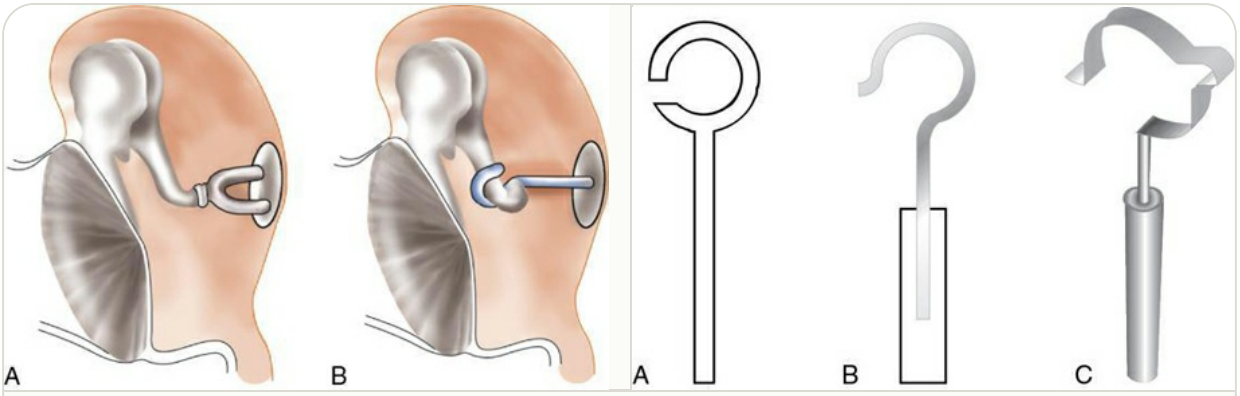


FIGURE 6 First, the fixed stapes before removal, and a Teflon piston in its place afterwards. Second, stapes prostheses: Teflon, platinum–Teflon and titanium–Teflon pistons.
Dhingra, figs 13.4 and 13.5

7.4.5 Results

The air-bone gap closes to within 10 dB in about 90 to 95% of experienced hands. Carhart's notch disappears with it.

OPERATIVE GUIDES IN THEATRE 3 operations

Otosclerosis
Stapedotomy
Bone-Anchored Hearing Devices
Bone-anchored devices
Cochlear Implantation in Adults
Implant for far-advanced disease

7.5 Far-advanced otosclerosis

When combined loss is so severe that aids no longer help, a cochlear implant gives good results. Stapes surgery plus a powerful aid is an alternative in selected patients. Otosclerotic bone can cause facial nerve stimulation by the implant.

7.6 Historical operations

Stapes mobilisation gave only temporary benefit, as the stapes refixed. Lempert's fenestration created a new window in the lateral semicircular canal; it left a mastoid cavity and an inherent loss of about 25 dB, and is now obsolete.

8 CHAPTER EIGHT Complications of stapes surgery

Rare, but the patient must hear about them beforehand.

8.1 During the operation

- **Tear of the tympanomeatal flap,** which can leave a perforation.
- **Chorda tympani injury,** causing a metallic taste in 5–10%, worse if the opposite chorda was already damaged.
- **Incus dislocation.**
- **Floating footplate,** which drops into the vestibule if pushed.
- **Perilymph gusher,** about 1 in 300, more likely with a wide cochlear aqueduct, especially in X-linked deafness.
- **Facial nerve injury,** more likely with a dehiscent or overhanging nerve.

8.2 Early after surgery

8.2.1 Vertigo

Usually from surgical trauma, serous labyrinthitis or a prosthesis that is too long. Persistent severe vertigo needs urgent review.

8.2.2 Reparative granuloma

Rising sensorineural loss and vertigo at one to two weeks; needs urgent re-exploration.

8.3 Late complications

8.3.1 Conductive loss returning

A short, loose or displaced prosthesis, or erosion of the incus (the commonest finding at revision, in about 5%).

8.3.2 Perilymph fistula

Fluctuating hearing and vertigo, often after straining or barotrauma.

8.3.3 Sensorineural loss

About 2% have some sensorineural loss; about 1 in 200 lose all hearing in the operated ear (a *dead ear*). A slow high-frequency loss may appear over the years.

AFTER STAPES SURGERY

1. Lie the patient with the operated ear up for the first hours; expect mild unsteadiness.
2. Vertigo with nystagmus, sudden hearing loss or facial weakness needs same-day senior review.
3. Advise against straining, nose blowing, heavy lifting and flying for about two to four weeks.
4. Avoid water in the ear until the flap has healed.
5. Book an audiogram at about six weeks.

NUMBERS AND CURRENT PRACTICE (OXFORD TEXTBOOK)

Clinical otosclerosis affects about 2% of adults, rising to 3% over 60, although histological foci are found in about 3.4% of temporal bones; the familial form (25 to 50%) is autosomal dominant with incomplete penetrance, and the measles link remains unproven. Population data show similar prevalence in men and women, with women more severely affected, which may explain the 2:1 female excess in surgical series. Untreated, the air–bone gap widens by about 1 dB a year, with a further 1 dB a year of sensorineural decline.

Hearing aids give excellent results and should be offered to every patient before surgery; a bone-conduction device suits those who cannot wear an aid and decline surgery. **Fluoride** is not used in the UK because its evidence is weak, and bisphosphonates remain experimental.

Small fenestra stapedotomy closes the air–bone gap to within 10 dB in 80% to over 95% of ears. Total sensorineural loss occurs in about 0.4 to 2%, taste disturbance from chorda tympani handling in up to 30% (usually temporary), and vertigo is now uncommon. Pre-existing tinnitus improves in about three-quarters and worsens in 5%. A delayed facial palsy after surgery usually recovers fully within six weeks on oral steroids.

Far-advanced otosclerosis (thresholds worse than 85 dB with unmeasurable bone conduction) is treated first by stapedotomy with a hearing aid, then by cochlear implantation if that fails; an active middle ear implant coupled to a stapedotomy piston suits patients too good for an implant.

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Site of predilection:** the fissula ante fenestram.
- **Paracusis Willisii,** and why it happens.
- **Schwartze sign:** active, vascular focus.
- **Carhart's notch:** 2000 Hz dip in bone conduction, mechanical, disappears after surgery.
- **Selection criteria and contraindications** for stapes surgery, especially the only hearing ear.
- **van der Hoeve syndrome:** osteogenesis imperfecta, blue sclerae, deafness.
- **Dead ear** risk of about 1 in 200, and gusher in X-linked deafness.

9.2 Common pitfalls

COMMON PITFALLS

- Operating on the only or better-hearing ear.
- Missing superior canal dehiscence: the reflex is present.
- Reading Carhart's notch as a true cochlear loss and turning a good candidate away.
- Forgetting to ask about occupation, flying and diving before offering surgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 20, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 13 and K. J. Lee ch. 19, 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · BONE DISEASE

Otosclerosis, Radiologic Signs around the Oval Niche

IMAGING FINDINGS

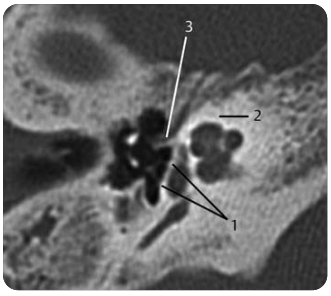
- Lucent focus at the fissula ante fenestram, anterior to the oval window
- Thickened footplate or obliterated oval window niche
- Cochlear form: pericochlear lucent halo ("double ring")

TOP DIFFERENTIALS

- Tympanosclerosis
- Osteogenesis imperfecta
- Paget disease of the temporal bone

CHECKLIST

- Compare both ears: usually bilateral
- Report round window involvement and cochlear disease before stapes surgery



CT, axial. At the edges of the footplate, two densities are visible (1), indicating osseous thickening, which may be considered as otosclerotic foci,...



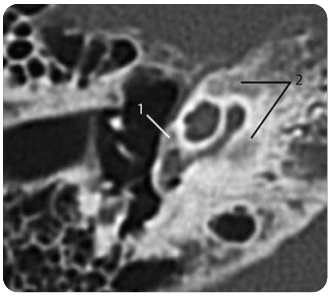
Coronal right temporal bone CT shows a lytic focus anterior to the oval window, the typical appearance and location of an otospongiotic plaque o...

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1120.

INNER EAR & IAC · BONE DISEASE

Otosclerosis, Retrofenestral

- Extension of demineralization. It may be unilateral or bilateral and may vary between both ears. Most cases of otosclerosis demonstrate minor deformities as described in Pathology of the Middle Ear earlier in this chapter.
- In cases of severe bilateral hearing disorders or complete deafness, cochlear implantation may be considered, although the patency of the cochlear lumen is essential for correct and complete electrode insertion. There is a risk of going down a false route due to the severe demineralization and distortion of the normal anatomy.

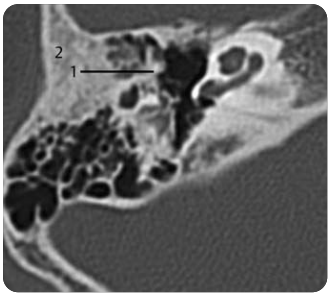


CT, axial. Fenestral otosclerosisas indicated by the lucent lesion at the fissula ante fenestram (1). Note the obviousretrofenestral otosclerosis as…

MIDDLE EAR & MASTOID · BONE DISEASE

Fixation of the Ossicular Chain

- As mentioned above in differential diagnosis, particular attention must be paid to patients with conductive hearing loss in combination with a normal appearing eardrum and aerated middle ear.
- In case of ankylosis of the ossicular chain, accurate evaluation of thin axial and coronal slices may reveal deformities and/or fixation.

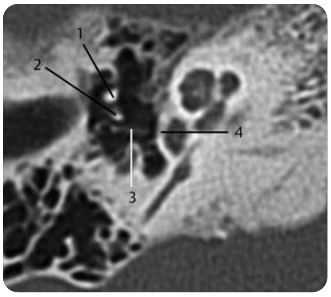


CT, axial. Although the conductive hearing loss might be ascribed to the atresia, the handle of the malleus (1) is fixed to the lateral bony wall of the…

MIDDLE EAR & MASTOID · AFTER SURGERY

Stapedectomy, Control of Piston Position

- Ossicular prostheses may differ as regards their degree of radiopacity and visibility on CT. For this reason fine slices are required.
- Malpositioning of the piston around the incus and possible erosions or resorbtion of the long process and lenticular process due to compression of the piston are difficult to evaluate on CT.

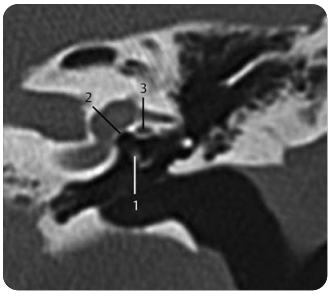


CT, axial. This view shows the handle of malleus (1) and posterior to it the long process of the incus (2), and a completely luxated pis- — ton (3) fro…

MIDDLE EAR & MASTOID · BONE DISEASE

Osteogenesis Imperfecta

- In osteogenesis imperfecta affecting the middle ear, fractures of the stapedial crus are often found on middle ear inspection and/or fixation of the footplate in combination with a known (family) history of osteogenesis imperfecta. Note the typical blue sclerae of the eyes in osteogenesis imperfecta type I.
- Extension of demineralization, which may be unilateral or bilateral and may vary between both ears. Most cases of osteogenesis imperfecta have minor abnormalities as described in Pathology of the Middle Ear earlier in this chapter. A clinical history of frequent and multiple fractures due to minor trauma everywhere in the body, as well as the typical blue sclerae, might help make the diagnosis.
- In cases of severe bilateral hearing disorders or complete deafness, cochlear implantation may be considered, although the patency of the cochlear lumen is essential for correct and complete electrode insertion. There is the risk of going down a false route due to the severe demineralization and distortion of the normal anatomy.

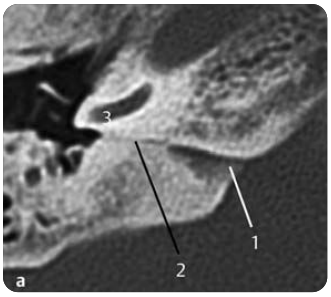


Patient with osteogenesis imperfecta and conductive hearing loss evaluated with CT, coronal. The radiologist explicitly described an enlarged…

INNER EAR & IAC · INFECTION & INFLAMMATION

Gusher

- A wide communication between the subarachnoid space and the perilymph (a defect at the fundus of the IAC or a dilated cochlear aqueduct) lets CSF pour out when the footplate is opened.
- CT clues: a bulbous, widened lateral IAC with a deficient fundus, absent modiolus and a corkscrew cochlea (X-linked incomplete partition type III, POU3F4).
- In a boy with a congenital mixed or conductive loss, review the CT for these signs before any stapes surgery.



CT, axial. At stapedotomy, abundant flow of inner ear fluid, the gusher phenomenon, made further surgical exploration impossible and the hole i…